

5	(a)	(electrostatic) force between two <u>point</u> charges <u>proportional</u> to <u>product of charges</u> and <u>inversely proportional</u> to the <u>square</u> of the distance (or separation)			B1
	(b)	(i)	field strengths (between spheres) are <u>opposite directions</u> because <i>either</i> <u>positive</u> (before 3.3 cm) and <u>negative</u> (after 3.3 cm) on graph <i>or</i> <u>$E = 0$ at a point</u> between spheres so, <u>same</u> sign		M1 A1
		(ii)	1.	$E = 3.0 \times 10^3 \text{ V m}^{-1}$ $a = qE / m = (1.60 \times 10^{-19} \times 3.0 \times 10^3) / (1.67 \times 10^{-27})$ $a = 2.9 \text{ (or } 2.87) \times 10^{11} \text{ m s}^{-2}$	C1 C1 A1
			2.	evidence of correct estimation of ΔV by area under graph between 3.3 to 5 cm $(1.60 \times 10^{-19}) \times \Delta V = \frac{1}{2} (1.67 \times 10^{-27}) v^2$ $6.03 \times 10^4 \text{ m s}^{-1} < v < 6.64 \times 10^4 \text{ m s}^{-1}$	M1 M1 A1
6	(a)				